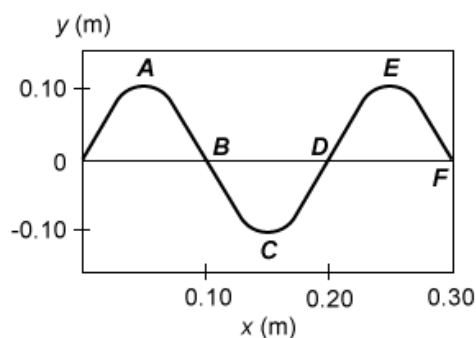


The following graph shows the vertical and horizontal displacement of an ocean wave at a given instant of time.



Between which two points would a horizontal measurement be equivalent to the wavelength of the wave?

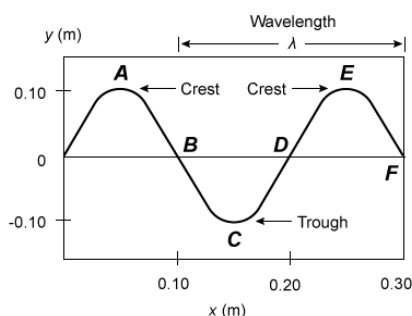
- ☐ A. Points *A* and *B* [1%]
- ✓ ☒ B. Points *B* and *F* [84%]
- ☐ C. Points *C* and *E* [6%]
- ☐ D. Points *D* and *F* [7%]

Correct

84% answered correctly

Explanation:

The measurement of wavelength (λ) in graphical representation of mechanical waves



Wavelength λ is the distance between two successive crests or troughs of a wave. In addition, the distance between two adjacent points with the same behavior is a valid measurement of wavelength.

In this question, the graph of an ocean wave is given, and the wavelength can be measured using the distance between adjacent crest (ie, the distance between *A* and *E*) or trough points. However, the distance between *A* and *E* is not available as an answer choice. Other horizontal distances between the points shown in the graph, equivalent to the distance between *A* and *E*, need to be measured.

The distance between *B* and *F* is the only horizontal distance equivalent to that of *A* and *E*. Furthermore, the behavior of the graph at those points is the same (ie, zero crossing with downward slope at both *B* and *F*). Therefore, the wavelength of the ocean wave is the distance between *B* and *F*.

(Choice A) The horizontal distance between *A* and *B* is one fourth of a full wavelength, which is measured crest-to-crest or trough-to-trough. The vertical distance between *A* and *B* is the amplitude of the wave.

(Choices C and D) The horizontal distances between *C* and *E*, and between *D* and *F*, are each half of a full wavelength. A full wavelength is measured crest-to-crest or trough-to-trough.

Educational objective:

Wavelength is the distance between successive crests or troughs of a wave. A horizontal measurement equal to that of the distance between successive crests or troughs is also a valid measurement of wavelength.

Time spent:0 sec

Subject:
Physics

QID:403995

System:
Light and Sound